

Lesson 9

Chapter 3 (continued): duality, Hicksian demand, the Slutsky matrix, and welfare evaluation

1 Utility maximization vs. expenditure minimization

The two problems are twins of each other. For $p \gg 0$, $w > 0$ and $\bar{u} > u(0)$,

$$e(p, v(p, w)) = w \quad \text{and} \quad v(p, e(p, \bar{u})) = \bar{u}.$$

The first relation says that the cheapest way to obtain the highest utility one can get with wealth w is going to cost exactly w . Otherwise, I have failed to solve one of the two problems correctly.

Important. Given prices, the two functions are *inverses* of each other: $e(p, \cdot)$ and $v(p, \cdot)$ undo each other. If I know one, then I can infer the other.

The two solutions coincide

For similar reasons, the solutions to the two problems coincide, in the sense that

$$h(p, \bar{u}) = x(p, e(p, \bar{u})) \quad \text{and} \quad x(p, w) = h(p, v(p, w)),$$

where $h(p, \bar{u})$, the solution to the expenditure minimization problem, is the *Hicksian demand*.

2 Hicksian (compensated) demand

The first equation also describes Hicksian demand, which is sometimes called *compensated demand*: when prices change, the consumer is compensated such that his wealth is just enough to achieve the same utility as before. This is also known as *Hicksian wealth compensation*.

Compare this to the *Slutsky* wealth compensation, which allows the consumer to buy the same *bundle* as before.

Assume the solution to the expenditure minimization problem is unique. Then h is a demand *function* rather than a demand correspondence.

The compensated law of demand

Hicksian demand also satisfies the compensated law of demand:

$$(p' - p) \cdot (h(p', \bar{u}) - h(p, \bar{u})) \leq 0.$$

Proof. An easy proof, by the definition of Hicksian demand. At prices p' the bundle $h(p', \bar{u})$ is the cheapest way to reach $\bar{u}$, and $h(p, \bar{u})$ also reaches $\bar{u}$, so

$$p' \cdot h(p', \bar{u}) \leq p' \cdot h(p, \bar{u}) \quad \Longleftrightarrow \quad p' \cdot (h(p', \bar{u}) - h(p, \bar{u})) \leq 0.$$

In the same way, at prices p ,

$$p \cdot h(p, \bar{u}) \leq p \cdot h(p', \bar{u}) \quad \Longleftrightarrow \quad p \cdot (h(p', \bar{u}) - h(p, \bar{u})) \geq 0.$$

Subtracting the second from the first, the compensated law of demand follows. □

3 Shephard's lemma and the Slutsky matrix

To further explore the links between the two versions of the compensated law of demand, start with *Shephard's lemma*:

$$h_i(p, \bar{u}) = \frac{\partial e(p, \bar{u})}{\partial p_i}.$$

Proof (by the envelope theorem). The expenditure minimization problem

$$\min_{x \geq 0} p \cdot x \quad \text{s.t.} \quad u(x) \geq \bar{u}$$

has the Lagrangian

$$\mathcal{L} = p \cdot x - \mu (u(x) - \bar{u}).$$

By the envelope theorem, the derivative of the value function $e(p, \bar{u})$ with respect to the parameter p_i is the partial derivative of the Lagrangian, evaluated at the optimum. Since p_i enters $\mathcal{L}$ only through the term $p_i x_i$,

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} \Big|_{\substack{x=h(p, \bar{u}) \\ \mu=\mu^*}} = h_i(p, \bar{u}).$$

Intuitively: when p_i rises a little, the first-order effect on the minimum cost is that the bundle already being bought, $h(p, \bar{u})$, costs $h_i(p, \bar{u})$ more per unit of price increase; re-optimizing the bundle has only a second-order effect. $\square$

An own-price change

Now consider an own-price change. Differentiating Shephard's lemma once more,

$$\frac{\partial h_i(p, \bar{u})}{\partial p_i} = \frac{\partial^2 e(p, \bar{u})}{\partial p_i^2} \leq 0,$$

since the expenditure function is concave in p . Hicksian demand is decreasing in its own price.

A cross-price change: the Slutsky equation

Consider next how the demand for good i depends on the price of good j . Start from $h_i(p, \bar{u}) = x_i(p, e(p, \bar{u}))$ and differentiate with respect to p_j ; by the chain rule,

$$\begin{aligned} \frac{\partial h_i(p, \bar{u})}{\partial p_j} &= \frac{\partial x_i(p, e(p, \bar{u}))}{\partial p_j} \\ &= \frac{\partial x_i(p, w)}{\partial p_j} + \frac{\partial x_i(p, w)}{\partial w} \frac{\partial e(p, \bar{u})}{\partial p_j} \\ &= \frac{\partial x_i(p, w)}{\partial p_j} + \frac{\partial x_i(p, w)}{\partial w} h_j(p, \bar{u}), \end{aligned}$$

with $w = e(p, \bar{u})$, using Shephard's lemma in the last step. Evaluated at $\bar{u} = v(p, w)$, we have $h_j(p, \bar{u}) = x_j(p, w)$, so

$$\frac{\partial h_i(p, \bar{u})}{\partial p_j} = \frac{\partial x_i(p, w)}{\partial p_j} + \frac{\partial x_i(p, w)}{\partial w} x_j(p, w).$$

This is the same expression as the entries of the *Slutsky substitution matrix*.

What we now know about the Slutsky matrix

So now we know that the terms in the Slutsky substitution matrix are of the form

$$\frac{\partial h_i(p, \bar{u})}{\partial p_j} \quad \text{or} \quad \frac{\partial^2 e(p, \bar{u})}{\partial p_i \partial p_j} :$$

the Slutsky matrix is just the Hessian of $e(p, \bar{u})$ in p .

- Since $e(p, \bar{u})$ is concave in p , this is a *negative semi-definite* matrix.
- The matrix is also *symmetric* (the cross partials of e do not depend on the order of differentiation). This is new, and not something we got out of WARP. However, SARP can be shown to give symmetry.

4 Welfare evaluations

What happens to the consumer's well-being if p or w changes? Focus on p .

The *qualitative* question is easy if $\succsim$ is known, since all we need to ask is whether utility increases or decreases. The indirect utility function, $v(p, w)$, is ideal for this purpose.

But there is no *quantitative* interpretation: given $\succsim$, $u(x)$ is not unique, so $v(p, w)$ is not unique either. We therefore develop some measures that have a quantitative meaning:

- first, a particular kind of indirect utility function;
- later, welfare measures tied to Hicksian demand, in contrast to Marshallian demand.

The money-metric indirect utility function

Start with $\succsim$. Pick any $u(x)$ that represents $\succsim$, and derive $v(p, w)$ and $e(p, \bar{u})$. Now we construct another one, the *money-metric indirect utility function*. Fix some arbitrary price vector $\bar{p} \gg 0$ and calculate

$$v_m(p, w \mid \bar{p}) = e(\bar{p}, v(p, w)).$$

- This is an indirect utility function: it maintains the same order, because the transformation on the right, $e(\bar{p}, \cdot)$, is monotone (increasing in the utility level).
- Additionally, it is the same regardless of which $u(x)$ representing $\succsim$ we started with.

Why. $v_m(p, w \mid \bar{p})$ is just the smallest amount of money needed, at prices $\bar{p}$, to buy a bundle as good as the best bundles under (p, w) . But this is determined by $\succsim$ and not by $u(x)$.

Transcription notes

- Notation follows Lesson 7: the original writes capitals throughout ($P, V(P, w), E(P, U), H(P, U), X(P, w)$); these are typeset as $p, v(p, w), e(p, \bar{u}), h(p, \bar{u}), x(p, w)$, and the utility level U as $\bar{u}$. The money-metric function, written $V_m(P, w \mid \bar{P})$, is $v_m(p, w \mid \bar{p})$.
- The title page reads “Lesson 8: Demand theory part” with no part number; the subtitle is written to match Lesson 7. The lesson is numbered 9 here, following the course schedule (Lesson 8 is the integration material).

- “the cheapest to obtain the highest utility to U can Get with wealth w is going to cost ex. w ” is read as *the cheapest way to obtain the highest utility one can get with wealth w is going to cost exactly w* . “solve two prob.” carries the insertion “one of” above it.
- Under $H(P, U)$ the original labels “Hicksian demand” and, under the second relation, “sol. exp min prob.”; the label is attached to h in the text. “also describes with hicksian” is read as *also describes Hicksian demand*.
- “a.k.a. hicksen compensation” carries “wealth” inserted under a caret: *Hicksian wealth compensation*. “Slutsky we. compen.” $\rightarrow$ *Slutsky wealth compensation*; “to lout same bundle” $\rightarrow$ *to buy the same bundle*.
- “law of compensate law of demand” $\rightarrow$ *compensated law of demand*. In the proof the second inequality is written $P \cdot (H(P, U) - H(P, U)) \geq 0$; the first H is read as $H(P', U)$, as the line before it requires. **Added:** the reason each inequality holds and the subtraction step that gives the result.
- Shephard’s lemma: the original reads “Proof by en. Therom. Try to prove it, claude.” **Added:** the envelope-theorem proof, with the Lagrangian of the EMP and a one-line intuition.
- **Corrected:** the own-price result is written $\partial^2 E / \partial P_i^2 < 0$; concavity gives only ≤ 0 , and the text uses $\leq$. “Hicksian demand $\downarrow$ in its own price” is typeset as *decreasing* (in the weak sense).
- **Corrected:** in the cross-price derivation the last term of the third line is written $H_i(P, U)$; by Shephard’s lemma $\partial E / \partial P_j = H_j(P, U)$. Likewise the final line ends with $x_i(P, w)$; it is $x_j(p, w)$, which is the Slutsky term. The left side of the first line is written $H_i(P, U) / \partial P_j$ without the ∂ in the numerator. **Added:** the condition $\bar{u} = v(p, w)$ under which $h_j = x_j$.
- “Hessian” of $E(P, U)$: written $\partial^2 E / \partial P_i \partial P_j$, typeset as $\partial^2 e / \partial p_i \partial p_j$. “not someth. got out of WARP” $\rightarrow$ *not something we got out of WARP*; “SARP can be shown to giving symmetry” $\rightarrow$ *to give symmetry*. **Added:** the reason the matrix is symmetric.
- “if P on w changes” $\rightarrow$ p or w . “a particular kinda indirect utility func.” $\rightarrow$ *kind of*. “Later, the welfare to hicksen demand and contrast to marshallian demand” is read as *welfare measures tied to Hicksian demand, in contrast to Marshallian demand*. “Pick any $U(x)$ that pre. $\succsim$ ” $\rightarrow$ *represents*; “Perive” $\rightarrow$ *derive*.
- “it maitains the same order” $\rightarrow$ *maintains*; “the transformation of the right” $\rightarrow$ *on the right*. **Added:** that $e(\bar{p}, \cdot)$ is the transformation and why it is monotone.
- The last sentence on page 8 reads “ $V_m(P, U \mid \bar{P})$ is just the smallest amount of money needed to But the best bundles under (P, w) ”: the U is read as w (as in the definition) and “But” as *buy*; “at prices $\bar{p}$ ” and “a bundle as good as” are added for precision.
- Pages of the original, for tracing back: page 1 title; page 2 the duality identities and $h = x(p, e)$; page 3 compensated demand, Hicksian vs. Slutsky compensation, the compensated law of demand and the start of its proof; page 4 the end of the proof, Shephard’s lemma and the own-price effect; page 5 the cross-price derivation (Slutsky equation); page 6 the Hessian, negative semi-definiteness, symmetry and the start of welfare evaluation; page 7 no quantitative meaning and the money-metric definition; page 8 its properties.